

Data Analytics Internship — Level 3

Task 3: Price Range vs. Online Delivery & Table Booking

Objective

This report examines whether a relationship exists between a restaurant's price range and its likelihood of offering online delivery and table booking, and determines whether higher-priced restaurants are more likely to offer these services — addressing the Level 3, Task 3 brief.

Dataset Overview

Metric	Value
Total restaurant records	9,551
Restaurants in 1 - Low Cost	4,444
Restaurants in 2 - Moderate	3,113
Restaurants in 3 - Expensive	1,408
Restaurants in 4 - Very Expensive	586

Service Availability by Price Range

Price Range	Online Delivery %	Table Booking %
1 - Low Cost	15.77%	0.02%
2 - Moderate	41.31%	7.68%
3 - Expensive	29.19%	45.74%
4 - Very Expensive	9.04%	46.76%

Online Delivery vs. Table Booking Availability by Price Range

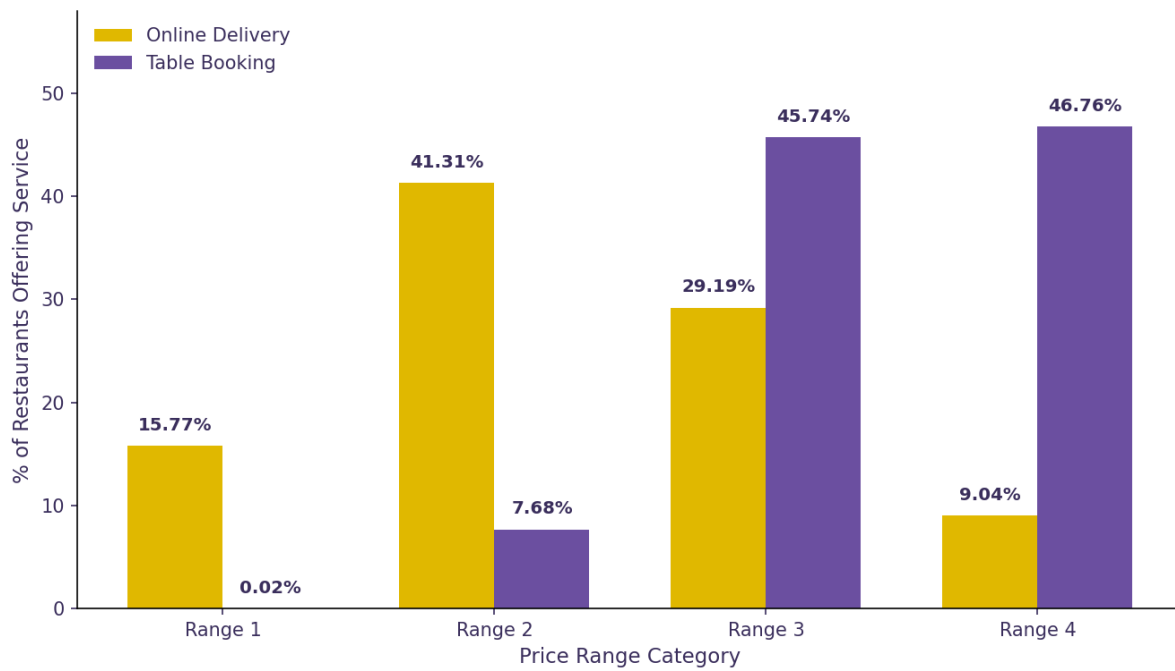


Figure 1: Online delivery and table booking availability compared across price ranges.

Trend Analysis

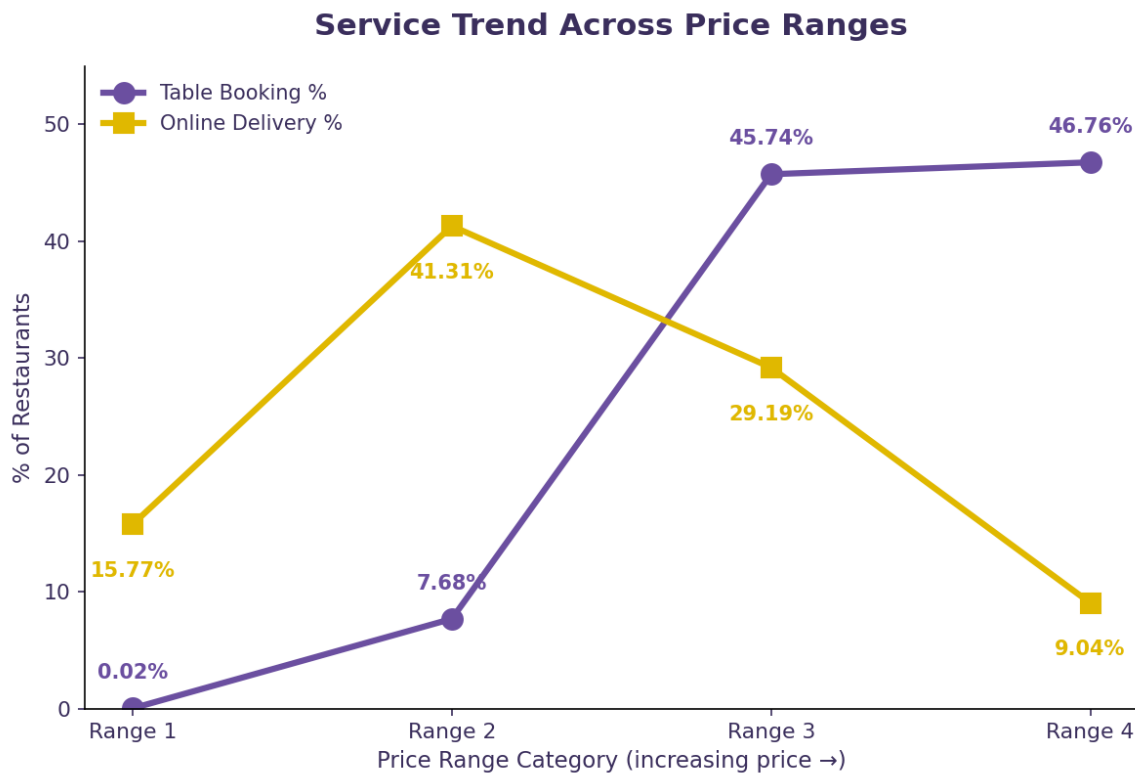


Figure 2: Table booking rises consistently with price; online delivery follows a different pattern.

Table booking: Rises steadily and consistently with price — from just 0.02% at the lowest price range to 46.76% at the highest. Higher-priced restaurants are clearly far more likely to offer table booking.

Online delivery: Does *not* follow the same pattern. It peaks at Price Range 2 (Moderate, 41.31%), then actually declines for Price Range 3 (29.19%) and drops further for Price Range 4 (9.04%, the lowest of all four categories). The most expensive restaurants are the least likely to offer online delivery.

Conclusion

The relationship between price range and service availability is not uniform across both services.

Higher-priced restaurants are clearly more likely to offer table booking — this holds true monotonically across all four price tiers, consistent with fine-dining establishments prioritizing reservation-based service. However, **higher-priced restaurants are not more likely to offer online delivery**; in fact, the cheapest and most expensive restaurants are both less likely to offer delivery than mid-range (moderate-priced) restaurants, which sit at the sweet spot for delivery-oriented business models. This suggests that price

range predicts table-booking availability well, but is a poor predictor of online delivery availability on its own.

Prepared for Cognifyz Technologies Data Analytics Internship — Level 3, Task 3